THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 6: **Perimeter and Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The area of a rhombus of side 25 cm and one diagonal 14 cm is (cm^2): 1

(a) 336 **(b)** 338 **(c)** 335 **(d)** 337

2. A triangle has sides 13 cm, 20 cm, 21 cm. The length of the shortest altitude (cm) is: 1

(a) 13 **(b)** 11 **(c)** 14 **(d)** 12

3. A rope of length 220 cm (take $\pi = \frac{22}{7}$) forms a circle. The same rope forms a square. The area of the square (cm^2) is: 1

(a) 3026 **(b)** 3024 **(c)** 3025 **(d)** 3027

4. A parallelogram and a triangle on the same base and between the same parallels have areas in the ratio: 1

(a) $2:1$ **(b)** $3:1$ **(c)** $1:2$ **(d)** $1:1$

5. The area of a cyclic quadrilateral with sides 6 cm, 14 cm, 18 cm and 22 cm is (cm^2): 1

(a) 192 **(b)** 194 **(c)** 191 **(d)** 193

6. A circle of radius 35 cm (take $\pi = \frac{22}{7}$) is inscribed in a square. The area of the square not covered by the circle is (cm^2): 1

(a) 1050 **(b)** 1052 **(c)** 1051 **(d)** 1049

7. Two concentric circles have radii 35 cm and 49 cm (take $\pi = \frac{22}{7}$). The area of the ring between them is (cm^2): 1

(a) 3698 **(b)** 3695 **(c)** 3696 **(d)** 3697

8. A wheel makes 200 revolutions to cover 616 m (take $\pi = \frac{22}{7}$). Its radius (cm) is: 1

(a) 51 **(b)** 48 **(c)** 50 **(d)** 49

9. **Assertion (A):** The length of the arc of a circle of radius 14 cm subtending 90° at the centre is 22 cm (take $\pi = \frac{22}{7}$). 1

Reason (R): Arc length $= \dfrac{\theta}{360} \times 2\pi r$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** By Brahmagupta's formula, the area of a cyclic quadrilateral with sides 6 cm, 8 cm, 6 cm, 8 cm is 48 cm^2. 1

Reason (R): A rectangle is a cyclic quadrilateral.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The area of a rhombus is 216 cm^2 and one diagonal is 24 cm. Find the other diagonal and the side of the rhombus. 2

12. Find the area of a cyclic quadrilateral with sides 6, 14, 18 and 22 cm. 2

13. A circle of radius 28 cm is inscribed in a square (take $\pi = \frac{22}{7}$). Find the area of the square left uncovered by the circle. 2

14. The area of a rhombus is 240 cm^2 and one diagonal is 30 cm. Find the other diagonal and the side of the rhombus. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A rope 220 cm long is first formed into a circle and then into a square (take $\pi = \frac{22}{7}$). Find the radius of the circle, the side of the square, and which encloses the larger area. 3

16. A running track is made of a rectangle 100 m $\times$ 56 m with a semicircle (radius 28 m) on each of the two shorter sides (take $\pi = \frac{22}{7}$). Find the length of the track along its boundary and the area enclosed. 3

17. The sides of a triangle are 12 cm, 17 cm and 25 cm. Find its area and the altitudes on the sides 12 cm and 17 cm. 3

OR

A rope 264 cm long is first formed into a circle and then into a square (take $\pi = \frac{22}{7}$). Find the radius of the circle, the side of the square, and which encloses the larger area. 3

18. A running track is made of a rectangle 50 m $\times$ 70 m with a semicircle (radius 35 m) on each of the two shorter sides (take $\pi = \frac{22}{7}$). Find the length of the track along its boundary and the area enclosed. 3

OR

The sides of a triangle are 25 cm, 25 cm and 14 cm. Find its area and the altitudes on the sides 25 cm and 25 cm. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A cyclic quadrilateral field has sides 25 m, 60 m, 52 m and 39 m, taken in order. 5

(i) Find its perimeter.

(ii) Find its semi-perimeter s.

(iii) Find the four values $s-a$, $s-b$, $s-c$, $s-d$.

(iv) Use Brahmagupta's formula to find the area of the field and the cost of sowing it at Rs 10 per m^2.

20. A park is a rectangle 90 m $\times$ 60 m. A circular flower bed of radius 21 m (take $\pi = \frac{22}{7}$) is in it, and a path 10 m wide runs round the park inside its boundary. Find the area of the park excluding the path, the area of the grass if the bed is removed from this, and the cost of turfing the grass at ₹2 per m^2. 5

OR

Two concentric circles have radii 28 cm and 35 cm (take $\pi = \frac{22}{7}$). Find the area of the ring and the difference between the two circumferences. 5

END OF QUESTION PAPER — Total: 40 Marks